

APPLIED MATHEMATICS

Important Question Bank & High-Frequency Topics

1. Matrices and Determinants High Weightage

- **Solving Systems of Linear Equations:** Mandatory 5-mark questions using the Matrix Method ($X = A^{-1}B$) or Cramer's Rule.
- **Matrix Algebra:** Finding constants (k) in equations like $A^2 - 8A + kI = 0$ and matrix inversion.

2. Financial Mathematics

- **EMI Calculations:** Flat Rate and Reducing Balance methods (Common case study).
- **CAGR:** Calculating growth rates for investments over specific time periods.
- **Linear Depreciation:** Annual depreciation, scrap value, and scheduling.
- **Perpetuity:** Present value calculations for payments at start/end of periods.

3. Time Series Analysis

- **Method of Least Squares:** Fitting straight-line trends for future forecasting.
- **Moving Averages:** Calculating 4-yearly or 5-yearly (centered) averages for data smoothing.

4. Calculus and its Applications

- **Maxima and Minima:** Profit maximization, cost minimization, and geometric optimization.
- **Cost/Revenue Analysis:** Mathematical relationships between MC and AC.
- **Differential Equations:** Finding order/degree and solving first-order equations.

5. Probability & Inferential Statistics

- **Poisson Distribution:** Probabilities, mean, and variance formulas.
- **t-Test:** Calculating Degrees of Freedom ($n_1 + n_2 - 2$) and Null Hypothesis testing.
- **Mathematical Expectation:** $E(X)$ and Variance from probability tables.

6. Numbers & Numerical Applications

- **Modulo Arithmetic:** Remainder of large powers and unit digit calculation.
- **Pipes and Cisterns:** Combined rates of inlet and outlet pipes.
- **Boats, Streams & Races:** Upstream/Downstream speeds and linear race dynamics.

7. Linear Programming (LPP)

- **Graphical Solutions:** Formulating inequalities and maximizing Objective Functions (Z) via corner points.